UNIVERSITY OF THE PEOPLE

BIOL 1301-01 - AY2026-T5

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Date: June 18, 2026

Applying the Scientific Method to Understand Difficulty Concentrating During Study Sessions

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Introduction

The scientific method is a systematic process that helps scientists investigate questions and solve problems using observation, evidence, and experimentation. While it is commonly used in scientific research, it can also be applied to everyday situations. One biological problem I have experienced is difficulty concentrating during long study sessions. As a university student, maintaining focus is essential for learning and academic success. By applying the scientific method, I can investigate possible causes of this problem and identify effective solutions.

Observation

Over the past several months, I have noticed that my concentration decreases significantly after studying continuously for approximately one to two hours. During these periods, I become distracted more easily, struggle to retain information, and feel mentally fatigued. This problem occurs most often when studying for demanding subjects such as programming, mathematics, or biology. I have also observed that the problem becomes worse when I sleep less than usual or spend extended periods using electronic devices before studying.

Based on these observations, the question I want to investigate is: Why is it difficult for me to concentrate during long study sessions?

Research

To better understand this issue, I reviewed information from biology resources and educational articles. Research indicates that adequate sleep plays a critical role in attention, memory formation, and cognitive performance. Sleep deprivation can impair concentration, decision-making, and learning abilities (Clark et al., 2020).

Studies also suggest that the human brain has a limited attention span during intensive mental activities. Taking short breaks can improve productivity and reduce mental fatigue. In addition, hydration and proper nutrition are important because the brain requires a constant supply of energy and nutrients to function effectively. Stress and excessive screen time may also contribute to reduced concentration and increased mental exhaustion.

After reviewing these findings, sleep quality appeared to be one of the most significant factors affecting concentration and academic performance.

Hypothesis

Based on the research, I propose the following hypothesis:

If I obtain at least seven to eight hours of sleep each night, then my ability to concentrate during study sessions will improve because sufficient sleep supports brain function, memory consolidation, and attention.

I believe this hypothesis is valid because scientific evidence consistently shows a relationship between sleep and cognitive performance. My personal observations also suggest that I perform better academically after a full night of sleep.

Applying the Scientific Method

To test this hypothesis, I will conduct a two-week investigation.

During the first week, I will follow my normal routine and record the number of hours I sleep each night. I will also measure my concentration by tracking how long I can study before becoming distracted and by recording my daily study productivity.

During the second week, I will intentionally maintain a consistent sleep schedule and ensure that I sleep between seven and eight hours each night. I will continue recording the same measurements used during the first week.

The steps of the experiment are as follows:

1. Record nightly sleep duration for two weeks.
2. Study at approximately the same time each day.
3. Measure the length of focused study sessions.
4. Record the number of distractions experienced during studying.
5. Track completion of daily study goals.
6. Compare the results from Week 1 and Week 2.

If concentration improves during the second week, the results will support the hypothesis. If there is no significant difference, then another factor such as nutrition, stress, hydration, or screen exposure may be responsible, and a new hypothesis can be developed and tested.

Conclusion

The scientific method provides a structured approach for investigating everyday biological questions. By observing a problem, conducting research, forming a hypothesis, and designing an experiment, it is possible to identify factors that influence concentration and learning. Applying this process to my study habits may help me improve academic performance while developing a better understanding of how biological processes affect human behavior.

References

Clark, M. A., Choi, J., & Douglas, M. (2020). *Biology 2e*. OpenStax. https://openstax.org/details/books/biology-2e

National Institute of Neurological Disorders and Stroke. (2025). *Brain basics: Understanding sleep*. https://www.ninds.nih.gov